Polynomials & Factoring Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Subtracting polynomials

ORIGINAL PROBLEM

$(5x^2 + 3x) - (2x^2 - 4x)$

INCORRECT WORKED SOLUTION

$(5x^2 + 3x) - (2x^2 - 4x)$

$= 5x^2 + 3x - 2x^2 - 4x$

$= 3x^2 - x$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Squaring a binomial

ORIGINAL PROBLEM

Expand $(x + 3)^2$

INCORRECT WORKED SOLUTION

$(x + 3)^2$

$= x^2 + 3^2$

$= x^2 + 9$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Multiplying binomials

ORIGINAL PROBLEM

Expand $(2x - 5)(x + 4)$

INCORRECT WORKED SOLUTION

$(2x - 5)(x + 4)$

$= 2x^2 + 8x - 5x + 20$

$= 2x^2 + 3x + 20$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Distributing a monomial

ORIGINAL PROBLEM

Expand $3x(x^2 + 2x - 4)$

INCORRECT WORKED SOLUTION

$3x(x^2 + 2x - 4)$

$= 3x^3 + 6x^2 - 4$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Factoring out the GCF

ORIGINAL PROBLEM

Factor $6x^2 + 9x$ completely.

INCORRECT WORKED SOLUTION

$6x^2 + 9x$

$= 3(2x^2 + 3x)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Factoring a trinomial

ORIGINAL PROBLEM

Factor $x^2 + 2x - 15$

INCORRECT WORKED SOLUTION

Find two numbers that multiply to -15 and add to 2: 5 and -3

$x^2 + 2x - 15 = (x - 5)(x + 3)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Difference of squares

ORIGINAL PROBLEM

Factor $x^2 - 49$

INCORRECT WORKED SOLUTION

$x^2 - 49$

$= (x - 7)^2$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Factoring when a ≠ 1

ORIGINAL PROBLEM

Factor $2x^2 + 7x + 3$

INCORRECT WORKED SOLUTION

$2x^2 + 7x + 3$

$= (2x + 3)(x + 1)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Zero product property

ORIGINAL PROBLEM

Solve $(x - 3)(x + 5) = 0$

INCORRECT WORKED SOLUTION

$(x - 3)(x + 5) = 0$

$x - 3 = 0$ or $x + 5 = 0$

$x = -3$ or $x = 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Setting equal to zero first

ORIGINAL PROBLEM

Solve $x^2 + 4x = 12$

INCORRECT WORKED SOLUTION

$x^2 + 4x = 12$

$x(x + 4) = 12$

$x = 12$ or $x + 4 = 12$

$x = 12$ or $x = 8$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points